

EMMA RAFKIN

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SUMMARY

NLP Research Scientist / Software Engineer with experience in applied AI and a focus in knowledge representation within LLMs. I am interested in making safer LLMs and expanding language technologies to speakers of all languages.

EDUCATION

Georgetown University

2023-2026

M.S. in Computational Linguistics

Overall GPA: 4.0

Master's Research Paper: Investigating the Cross-Lingual Transferability of NLP Tasks

Dartmouth College

2017-2021

B.A. in Computer Science

Minor in Linguistics

Overall GPA: 3.74, Cum Laude

EXPERIENCE

Johns Hopkins University Applied Physics Laboratory

August 2021 - Present

NLP Research Scientist / Software Engineer

Laurel, MD

- NLP subject matter expert and advisor for U.S. Government contractor. Create and deploy tools that inject domain knowledge into language models and extract critical information from text using state of the art NLP methods and domain ontologies
- Work on every part of the Applied AI pipeline: curate data and finetune LLMs to align with critical low-resource missions, create robust information retrieval and extraction systems, steer models during generation, construct thorough output validation and safety mechanisms
- Awarded 8 grants to pursue research in AI transparency, concept erasure, summarization, entity disambiguation, LLM domain adaptation, AI red teaming, and multi-modal anomaly detection
- Lead research teams of 3-6 people, present results to internal and external stakeholders
- Build software that can scale to process publicly available data streams of millions of documents per day

Dartmouth Applied Learning and Innovation (DALI) Laboratory

January 2019 - June 2021

Software Developer

Hanover, NH

- Produced full-stack web and mobile applications from scratch for external stakeholders every 10 weeks
- Created curricula and led classes in introductory web design and development for 30 students
- Developer mentor, Women in Science Program lead, core executive board member for 2020-2021

Alarm.com

June 2020 - August 2020

Software Engineer Intern

Tysons, VA

- Created automated full-stack client-facing reports for property managers about all devices in their units

Cambly Inc

January 2020 - March 2020

Software Engineer Intern

San Francisco, CA

- Upgraded Android application in Java and Kotlin to include English as a Second Language courses for adults

Dartmouth College Computer Science Department

September 2018 - December 2018

Teaching Assistant

Hanover, NH

- Taught weekly review class for Object Oriented Programming course to 10-15 students, graded coursework and tests

Thayer School of Engineering

IT Help Desk Technician

January 2018 - June 2018

Hanover, NH

- Answered computing questions for the students and faculty at Thayer
- Worked with clients to solve hardware difficulties. Implemented an organizational system across the computing services at the school

FEATURED RESEARCH PROJECTS

Dandelion

Mechanistic Interpretability

JHU/APL

In progress

- Identifying mission-relevant negative LLM behavior during runtime, steering in activation & weight space to prevent that behavior from surfacing
- PI for a team of 3 researchers, won a year-long internal research grant

Using Task Arithmetic for Cross-Lingual Transfer in LLMs

Cross-lingual Task Transfer

Georgetown University

Under Review

- Explores the cross-lingual transfer of form & meaning tasks in LLMs as distance increases between source and target languages

Task Arithmetic with Support Languages for Low-Resource ASR

Low-Resource Language Modeling

Georgetown University

Awaiting Publication

- Incorporates information from genetically related high-resource languages to improve low-resource ASR models
- Preprint on ArXiv, awaiting publication in ACL Anthology

CaTE

Trustworthy AI development

JHU/APL

Complete

- Guidance for software engineers and project managers on data curation methods for trustworthy AI development
- Full report on ArXiv, separate paper on domain elicitation techniques published in 2024 International Conference on Assured Autonomy (ICAA).

TECHNICAL SKILLS

ML/NLP

PyTorch, Hugging Face, fine-tuning & LoRA, activation probing & steering, RAG, Slurm

Software Languages

Python, JavaScript, TypeScript, React, Java, Kotlin

Databases

SQLite, PostgreSQL, Elasticsearch, MongoDB

CI/CD

Git, Docker

Natural Languages

English (native), Spanish (conversational), ASL (beginner)

RELEVANT COURSEWORK

Computer Science

Object-Oriented Programming

Algorithms

Digital Electronics

Databases

Watson

Human-Computer Interaction

Software Design & Implementation

Linear Algebra

Senior Design & Implementation Project

Linguistics

Semantics I (formal)

Syntax I

ASL I

Introduction to Linguistics

Sociolinguistics

Phonology

Psycholinguistics

Semantics (grammatical)

Language Gender & Sexuality

NLP

Corpus Linguistics

ML for NLP

Information Theory

Signal Processing

Empirical Methods in NLP

Ethics in CL and AI

CL Research Methods

Machine Learning

Computational Linguistics